Poster Abstract: Enhancing Fault Resilience of Air Quality Monitoring in San Joaquin Valley: A Data Equity Analysis

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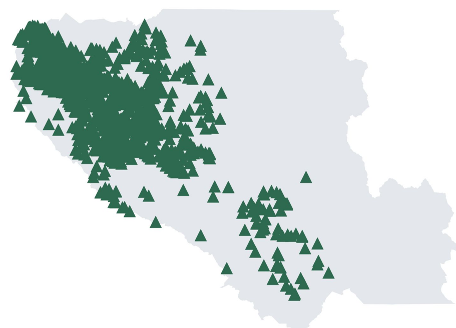
UCMERCED

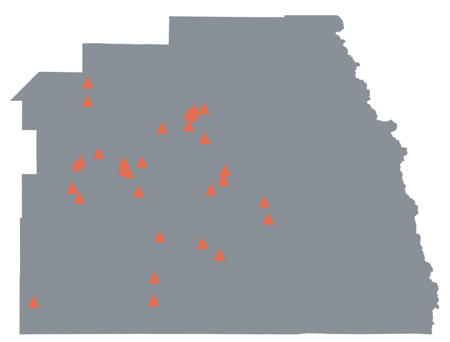
Motivation

Santa Clara County



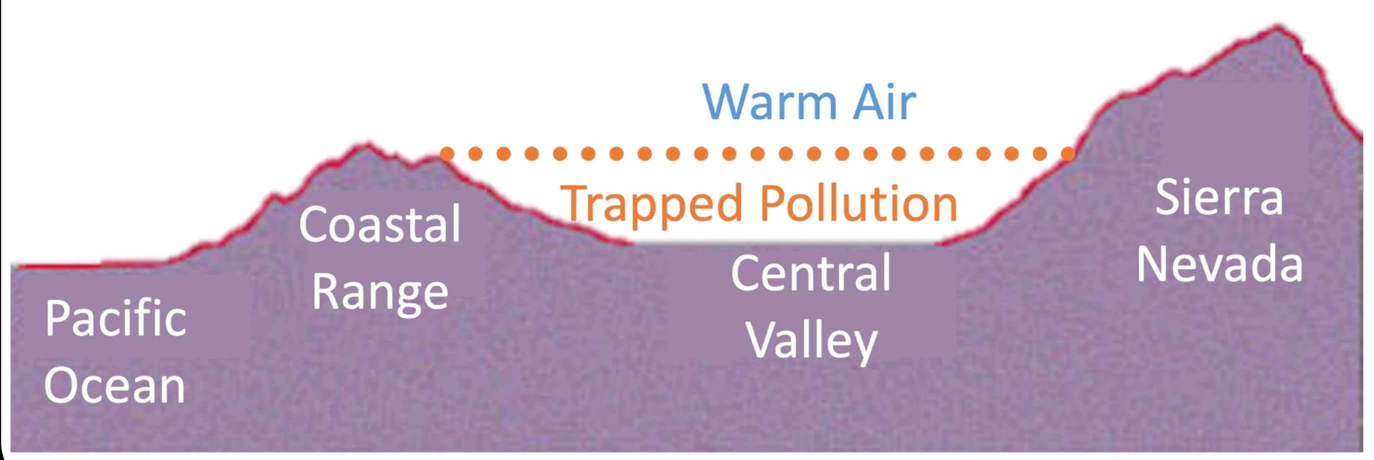
- Size: 3300km^2
- Population: 1871k
- Household Annual Income: 140258$
- # Air Quality Sensors: 1478

Tulare County



- Size:12490km^2
- Population:478k
- Household Annual Income:57394$
- # Air Quality Sensors: 38

San Joaquin Valley Has the Worst Air Quality in the Country



- Geography
- Land usage
  - Agriculture
  - Warehouse
- Wirefire

San Joaquin Valley Socio-Economic Challenges

- Poverty: 14 % residents live in poverty
- Health: extreme non-compliance with pollution standards (EPA)

Socio-Economic Inequity Leads to Data Inequity

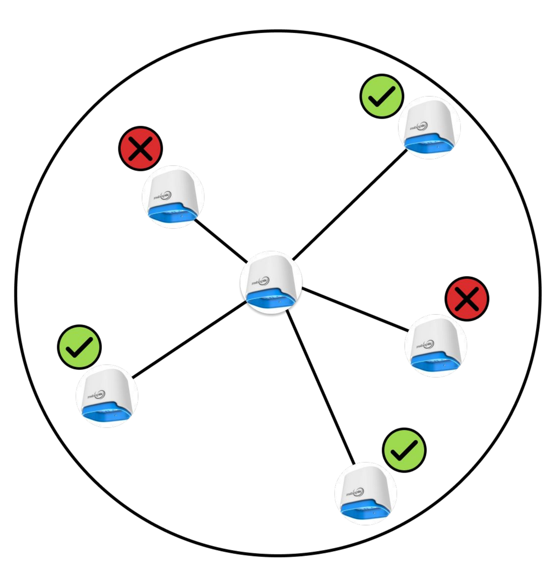
- Air Quality Monitoring System Fault Resilience
- Health Impacts Awareness

Question:
How do we quantify the impact of this inequity?

Data Inequity and Fault Resilience

- **Spatial Interpolation:** Use surrounding sensors' reading to interpolate the sensor's reading of target locations
- **Fault Resilience Measurement:** When a proportion of surrounding sensors is not available, how does the interpolation accuracy change?

Sensor Fault



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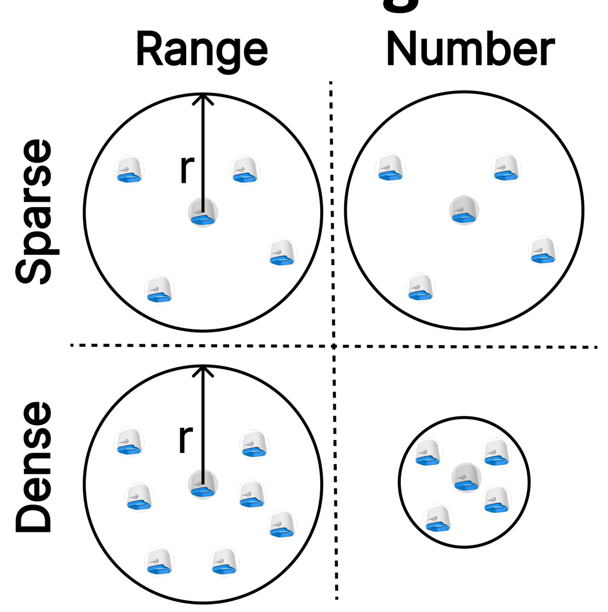
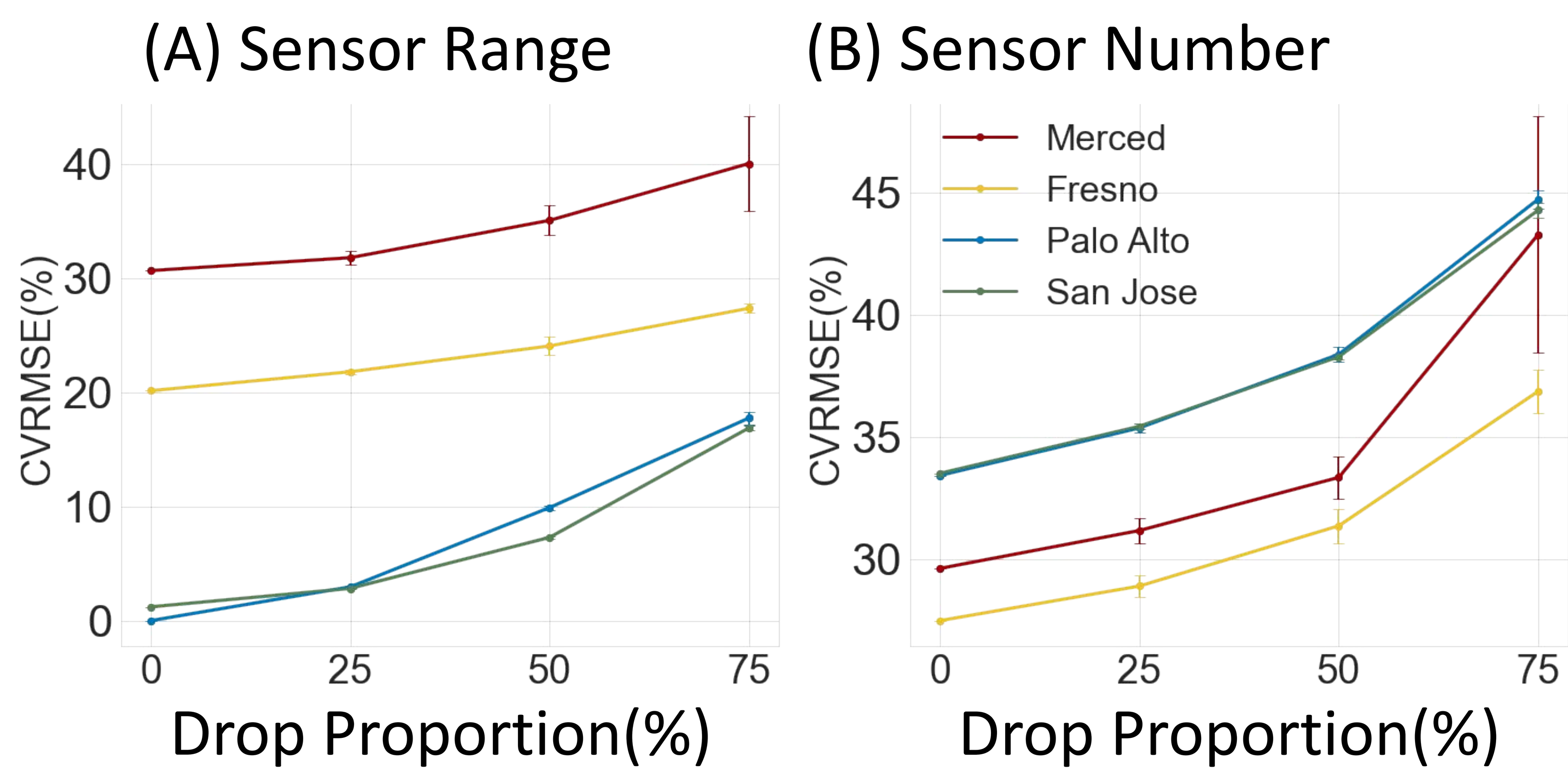
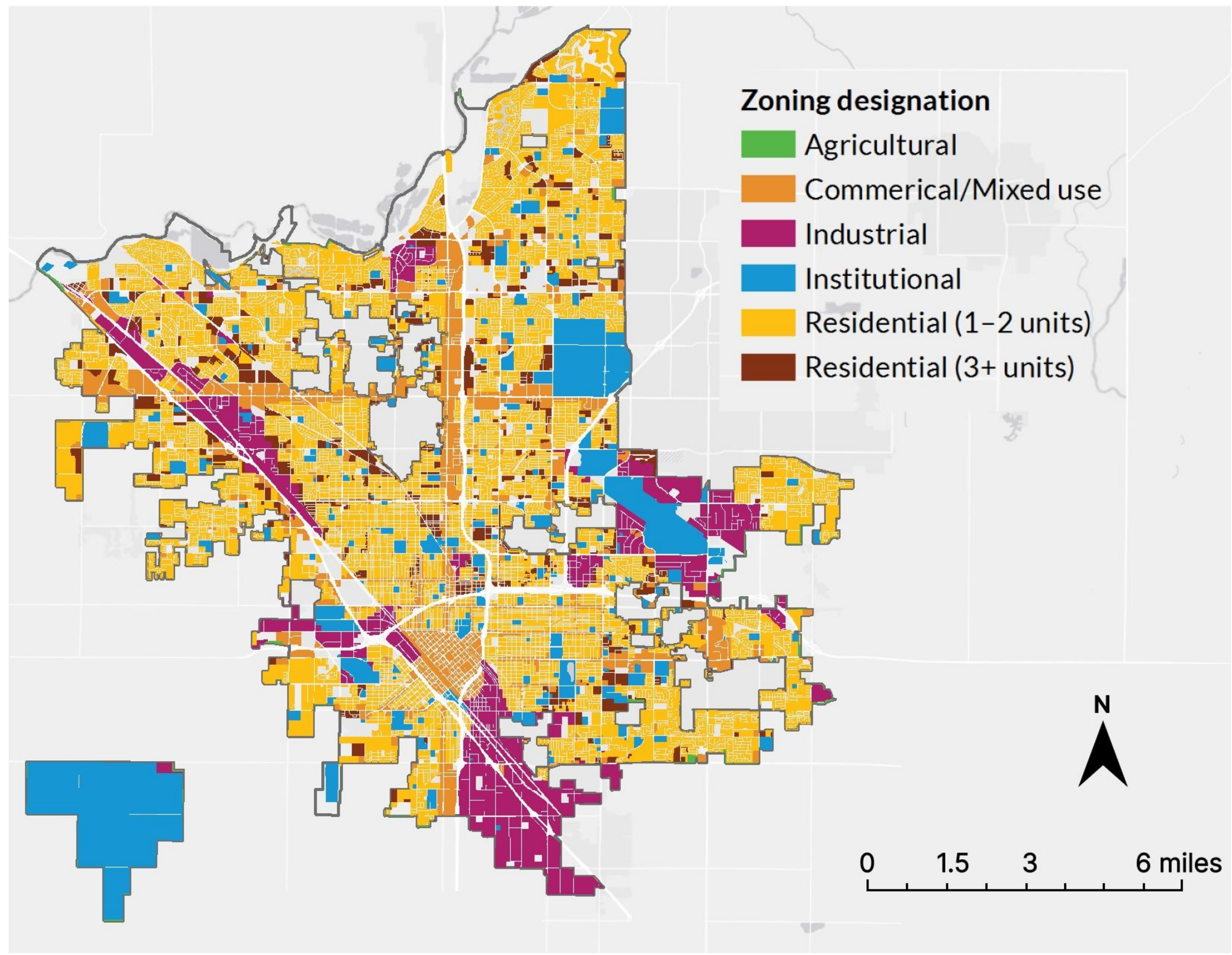




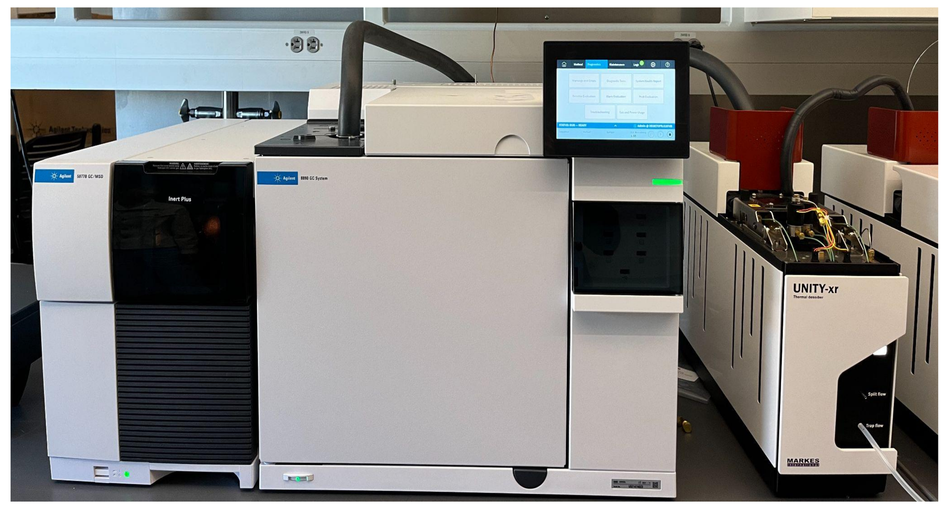
- Sparsity results in lower resilience
- Interpolation accuracy at Bay Area with 25% sensors outperforms that of SJV with 100% sensors
- Significant data inequity between SJV and Bay Area

Future Work

- Incorporating Zoning and Land Use Information for Modeling



- Decoupling PM2.5 Contributors
  - ❑ Chemical Compounds Analyzer
  - ❑ Cost: > $200k
  - ❑ Sample Period: > 2 days



  - ❑ PM2.5 Sensor
  - ❑ Cost: < $100
  - ❑ Sample Period: < 1 min